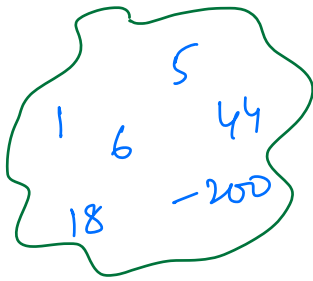


HashMap $\rightarrow$ Key-Value pairs

HashSet



```
HashSet<Integer> hs = new HashSet<>();
```

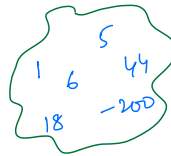
```
hs.add(1);
```

```
hs.add(5);
```

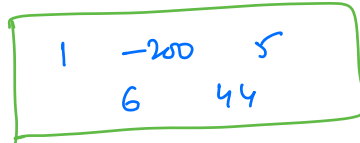
```
i
```

```
hs.add(44);
```

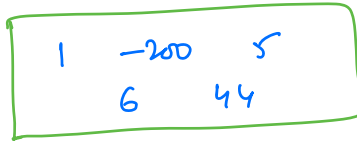
```
// Remove 18
```



```
hs.remove(18);
```



```
hs.add(6);
```



```
if (hs.contains(15))  $\rightarrow$  false
```

```
if (hs.contains(6))  $\rightarrow$  True  
System.out.println("Yay");
```

```
hs.size();  $\rightarrow$  5
```

```
for (int x : hs) {  
    System.out.println(x);  
}
```

ArrayList

Sequential order

Allows duplicates

HashSet

Sequence not maintained

Only unique elements present

```
int ar[] = ...
```

```
...
```

```
HashSet<Integer> hs = new HashSet<>();
```

```
for (int i = 0; i < ar.length; i++) {
```

```
    hs.add(ar[i]);
```

```
}
```

```
return hs;
```

Q) Given 2 Hashsets as input, print their common elements.

HS1 $\rightarrow \{0, -2, 3, 10\}$
HS2 $\rightarrow \{3, 6, -2, 4\}$ } -2, 3

```
for (int x: hs1) {  
    if (hs2.contains(x))  
        System.out.println(x);  
}
```

Q) Given a string, print the count of uppercase char.

A to Z $\rightarrow$ 65 to 90

```
int cnt = 0;  
for (int i = 0; i < str.length(); i++) {  
    char ch = str.charAt(i);  
    if (ch >= 65 && ch <= 90) OR if (ch >= 'A' && ch <= 'Z')  
        cnt++;  
}
```

Q) Given a string, print the count of special characters.

```
int cnt = 0;  
for (int i = 0; i < str.length(); i++) {  
    char ch = str.charAt(i);  
    if (! (ch >= 'A' && ch <= 'Z')  
        && ! (ch >= 'a' && ch <= 'z')  
        && ! (ch >= '0' && ch <= '9'))  
        cnt++;  
}
```

[Break till 10:28 PM]

```
StringBuilder sb = new StringBuilder();  
    ("");  
    ("apple");
```

```
sb.append("x"); //x  
sb.append("ab"); //xab  
sb.append('y'); //xaby  
sb.insert(2, "hq"); //xahqby  
sb.setCharAt(3, 'c'); //xahcby
```

```
String s = sb.toString();
```

